

Social Media Big Data Analytics

NTI Graduation Project • Big Data Analysis Track



Project Team

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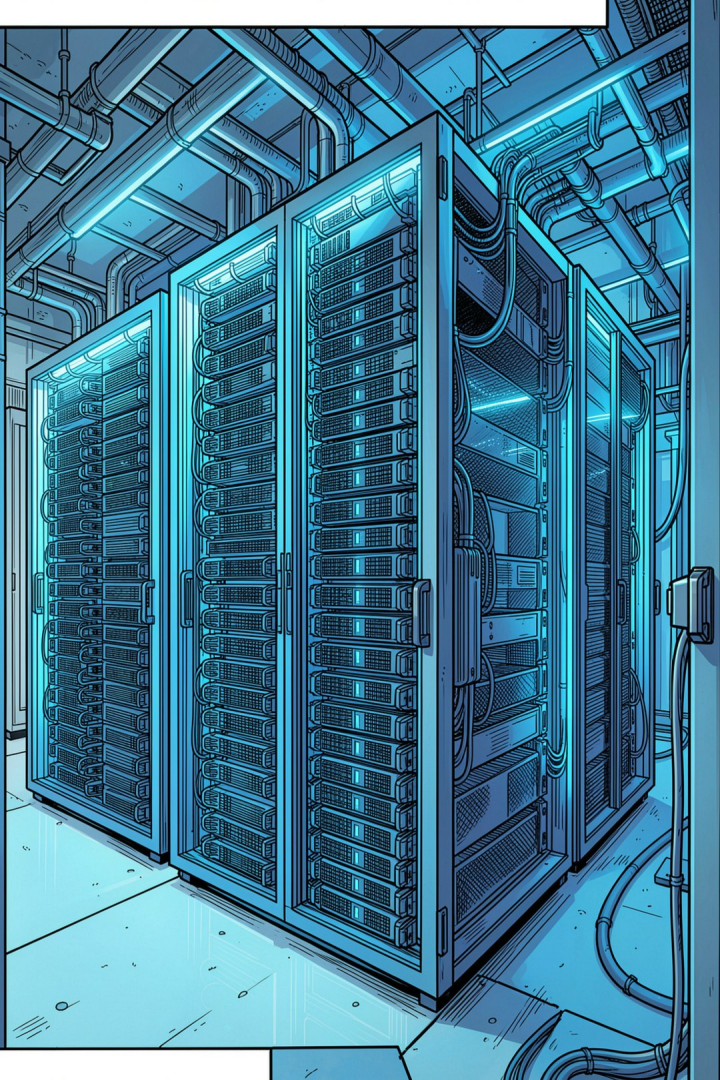
Fatma Abd Elnaser

Nourhan Anwar



Business Challenges

- Data Silos: cross-platform data fragmented and unstandardized
- Volume & Velocity: massive interactions require scalable architecture
- Unstructured Metadata: varied formats across text, video, image posts
- Actionable Insights Gap: hard to turn raw logs into strategic decisions



Project Objectives

Unified Data Warehouse

Build end-to-end pipeline using
Apache Hadoop and Hive SQL

Automated Aggregation

Structure unstructured social
metrics into normalized schemas

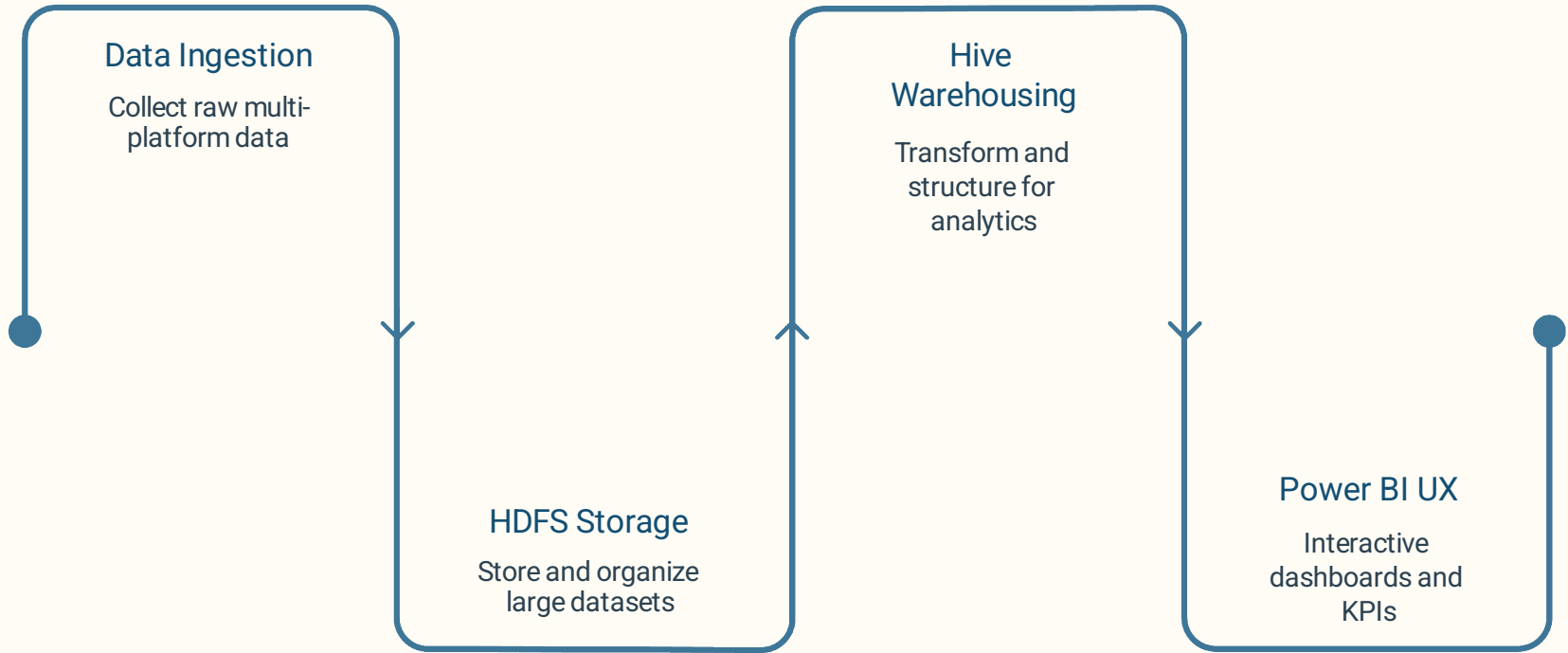
BI Visualization

Connect Hive backend directly to
interactive Power BI dashboards

Strategic Value

Empower marketing teams with
engagement and ROI insights

Pipeline Stages



Clear progression from raw multi-platform extraction to interactive executive dashboards and KPI analytics.



Stage 01 – Data Ingestion

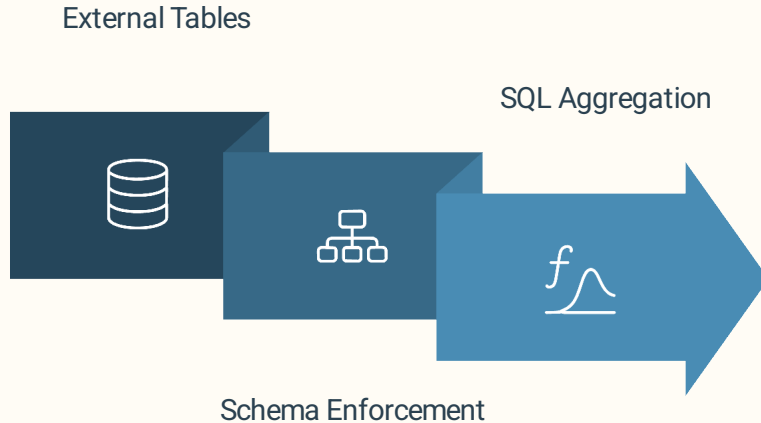
Raw FB/IG/YT data extracted and prepared into CSV/JSON formats.

Stage 02 – HDFS Storage

Distributed upload into Hadoop HDFS for fault tolerance and partitioned directories for performance.

Hive Warehousing

External tables over HDFS, schema-on-read flexibility, and optimized aggregation queries (grouped sum/avg).



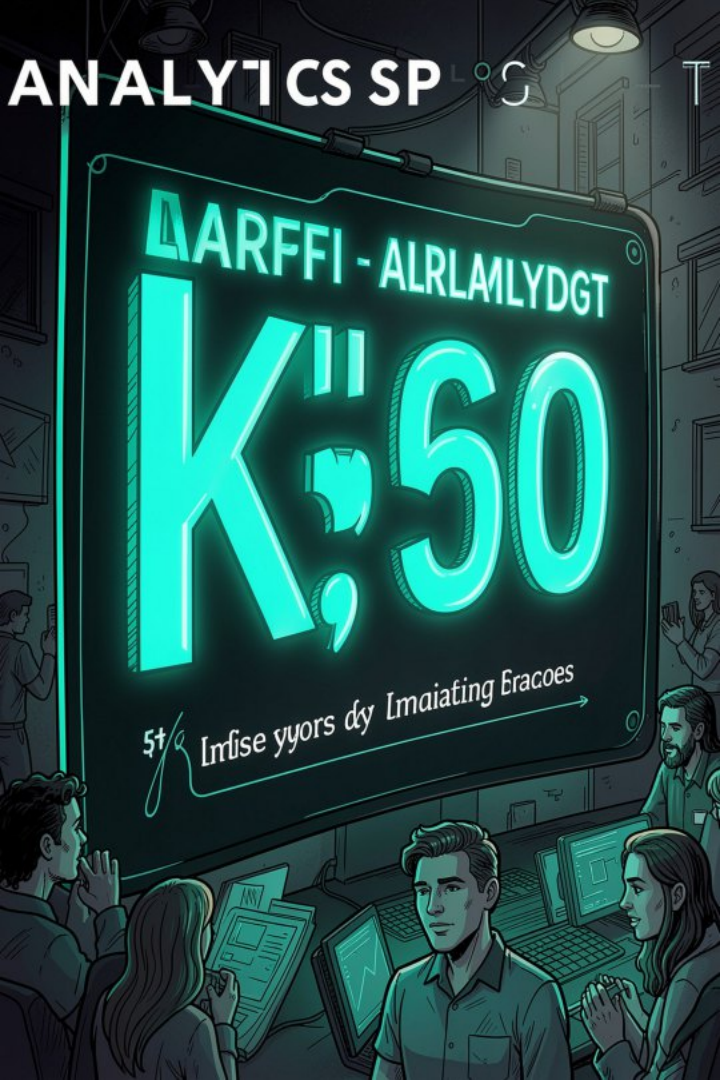
```
CREATE TABLE IF NOT EXISTS youtube_raw (  
  video_id STRING,  
  title STRING,  
  channel_title STRING,  
  category_id INT,  
  views BIGINT,  
  likes BIGINT,  
  dislikes BIGINT,  
  comment_count BIGINT  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
STORED AS TEXTFILE;
```



Power BI

ODBC integration delivers interactive dashboards with date range slicers, cross-filtering, and drill-downs to daily logs.

- Executive overview KPI tiles
- Dynamic filters & tooltips for granular metrics
- Custom navigation for overview → deep-dive workflows



Key Metrics & Insights

12.4M Impressions

Total impressions across FB, IG, YT

1.85M Engagements

Total engagements recorded

14.9% Avg Engagement Rate

Cross-platform average engagement

3 Platforms

Facebook, Instagram, YouTube



Strategic Takeaways & Next Steps

Content Strategy

Shift Instagram toward Reels & Carousels; leverage Shorts as top-funnel teasers to drive traffic to long-form YouTube content.

Scheduling & Optimization

Align Facebook posts to peak hours (7–10 PM) to increase organic distribution; favor evening scheduling for higher reach.

Measurement & UI

Surface high-priority KPIs in dark-themed executive dashboards with drill-downs, tooltips, and platform-specific color consistency.

Next Steps

Iterate ETL for finer partitions, expand ODBC tuning for real-time sync, and run A/B tests on Reels vs static posts to quantify uplift.

Thank you — Questions & Discussion

NTI BIG ATA ANALYTICS TRACK